

Kunal Avghade

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Summary

Backend-focused Software Engineer with **3+ years** designing highly scalable, fault-tolerant distributed systems. Proven track record of architecting petabyte-scale migration platforms and AI-driven workflows. Specialist in **Java/Spring Boot, Python/FastAPI, and TypeScript/NestJS** with strong algorithmic foundations and a passion for system design.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C++, SQL

Frameworks: Spring Boot, NestJS, FastAPI, React.js, Next.js

Distributed Systems: Temporal, Kafka, RabbitMQ, gRPC, REST

Cloud/DevOps: AWS, GCP, Docker, Kubernetes, Jenkins, ArgoCD

Databases: PostgreSQL, Redis, MongoDB, Milvus

Professional Experience

Software Development Engineer - II | Calfus Inc.

Jul 2023 – Present

Full Stack Development & Cloud-Native Engineering

Pune, India

• Aetherion — Agentic AI Platform

Tech Stack: Python, TypeScript, Java, FastAPI, Temporal, PostgreSQL, AWS, Docker, GenAI, LLMs, RAG

- Developed three distinct interfaces for agent creation—**Chat, LCNC (Low-Code/No-Code), and SDKs (Python, TS, Java)**—accelerating integration time and dynamically translating natural language into executable workflows via **LLMs and RAG**, leveraging **Milvus** for vector retrieval.
- Architected distributed orchestration using **Temporal**, guaranteeing fault-tolerant execution of long-running workflows and enabling the platform to seamlessly run **10,000+ agents in parallel**.
- Optimized asynchronous task pipelines, tool invocation layers, and **LLM token consumption**, significantly lowering execution latency, scaling throughput for concurrent, multi-tenant agent runs, and **reducing operational costs by 50%**.

• NetApp Data Migrator — ONTAP Storage Migration Platform

Tech Stack: TypeScript, NestJS, Java, Spring Boot, Temporal, gRPC, RabbitMQ, Redis, PostgreSQL, Keycloak, GCP, AWS, Azure, ONTAP

- Designed a scalable distributed migration platform for **NetApp ONTAP** capable of handling migrations up to **1+ PB** and **1+ Billion** files across **NFS/SMB** environments. Successfully migrated **100+ TB** and **90+ Million** files of unstructured data within **30 hours** with zero data loss during performance validation.
- Architected resumable, fault-tolerant migration workflows using **Temporal** across distributed worker clusters, supporting automated checkpointing and dead-letter recovery.
- Implemented highly parallel discovery engines and low-latency task coordination using **gRPC, RabbitMQ, and Redis** for dynamic workload partitioning, achieving **7x faster performance** compared to traditional Robocopy and rsync.
- Integrated **Keycloak** RBAC and built real-time operational telemetry to monitor cluster throughput, file-level checksum validation, and active worker health, while generating optimal **Chain of Custody (CoC) reports** at runtime.

• Asurint — Background Verification Microservices

Tech Stack: Java, Spring Boot, GCP, Kafka, ArgoCD, Jenkins, PostgreSQL

- Scaled **GCP**-based microservices for automated background check workflows, processing **millions of monthly indexing and verification requests** while ensuring **99% uptime**.
- Decoupled heavy PDF report generation from data ingestion pipelines using **Apache Kafka** event-driven messaging, eliminating API timeouts and reducing report generation latency.

Key Project

Vision-Based Intelligent Recipe Recommender

Tech Stack: Python, Django, YOLOv8, OpenCV, React, PostgreSQL, JWT

Jan 2023 – May 2023

- Trained a custom **YOLOv8** model to detect food ingredients from raw images, achieving **92% mAP**.
- Built a secure **Django REST API** and **React.js** client with **JWT**, mapping detected items to 5,000+ recipes via optimized SQL joins.

Achievements & Certifications

- Competitive Programming: LeetCode Guardian** (Max Rating: 2164, Top 1% globally); Solved **1000+** DSA problems across platforms. | LeetCode | Codeforces
- Drive Dynamo Award:** Won **3 consecutive years** (2023, 2024, 2025) for exceptional technical delivery and leading platform adoption at Calfus Inc.
- AWS Certified Solutions Architect – Associate & AWS Cloud Practitioner.**
- IBM AI Engineering Professional & Stanford Machine Learning Specialization.**

Education

Vishwakarma Institute of Information Technology

B.Tech in Electronics & Telecommunication (Software Systems Focus)

Pune, India

2020 – 2023